

Collaborative Activity: Integration by Parts

Name: _____

A. Find your partner

Find a student whose card matches yours: one card shows an integral and the other shows a choice of u and dv . Check that $u dv$ reproduces the integrand and differential exactly. Introduce yourselves, then complete the setup below. **Do not evaluate the integral yet.** Several cards are identical; any mathematically correct match is welcome.

Your pair's integral: _____

$u =$ _____ $dv =$ _____

$du =$ _____ $v =$ _____

Discuss with your partner: What will happen after one application of integration by parts? Will the new integral be simpler, or might we need to use the method again?

B. Definite integrals

Work with your partner.

$$\boxed{\int_a^b u dv = [uv]_a^b - \int_a^b v du}$$

$$\text{Recall: } \int a^x dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1).$$

Evaluate $\int_0^1 x 3^x dx$.

C. When once is not enough

Your TA will assign your pair **Problem 1** or **Problem 2**. Work together on that problem first. Prepare to explain why another application of integration by parts is needed. If you finish early, try the other problem.

Problem 1. Repeated integration by parts

Evaluate $\int x^2 e^{2x} dx$. Differentiate your final answer to check it.

$$u =$$

$$dv =$$

$$du =$$

$$v =$$

$$\bar{u} =$$

$$d\bar{v} =$$

$$d\bar{u} =$$

$$\bar{v} =$$

Problem 2. When the original integral returns

Evaluate $\int e^{2x} \sin x \, dx$.

$$u =$$

$$dv =$$

$$du =$$

$$v =$$

$$\bar{u} =$$

$$d\bar{v} =$$

$$d\bar{u} =$$

$$\bar{v} =$$

Compare with another pair

Join a pair that worked on the *other* problem. Each pair explains its solution while the other pair checks the signs and constant factors.

Discuss: In Problem 1, what becomes simpler each time? In Problem 2, why is seeing the original integral again useful rather than a reason to start over?

D. Optional extension: analyze an incorrect solution

A student writes

$$u = x, \quad dv = \sin(2x) dx, \quad du = dx, \quad v = \frac{1}{2} \cos(2x),$$

$$\int x \sin(2x) dx = \frac{x}{2} \cos(2x) - \frac{1}{2} \int \cos(2x) dx = \boxed{\frac{x}{2} \cos(2x) - \frac{1}{4} \sin(2x) + C}.$$

1. Differentiate the proposed answer to test whether it is correct.
2. Locate the *first* error and explain it.
3. Write a correct solution.